

SMART MULTI-MODAL TRANSPORTATION ROUTING

A Personalized, Energy-Efficient, and Environment-Aware Framework for Indian Cities

SOHAM SRIVASTAVA | ROHIT CHAUDHARI

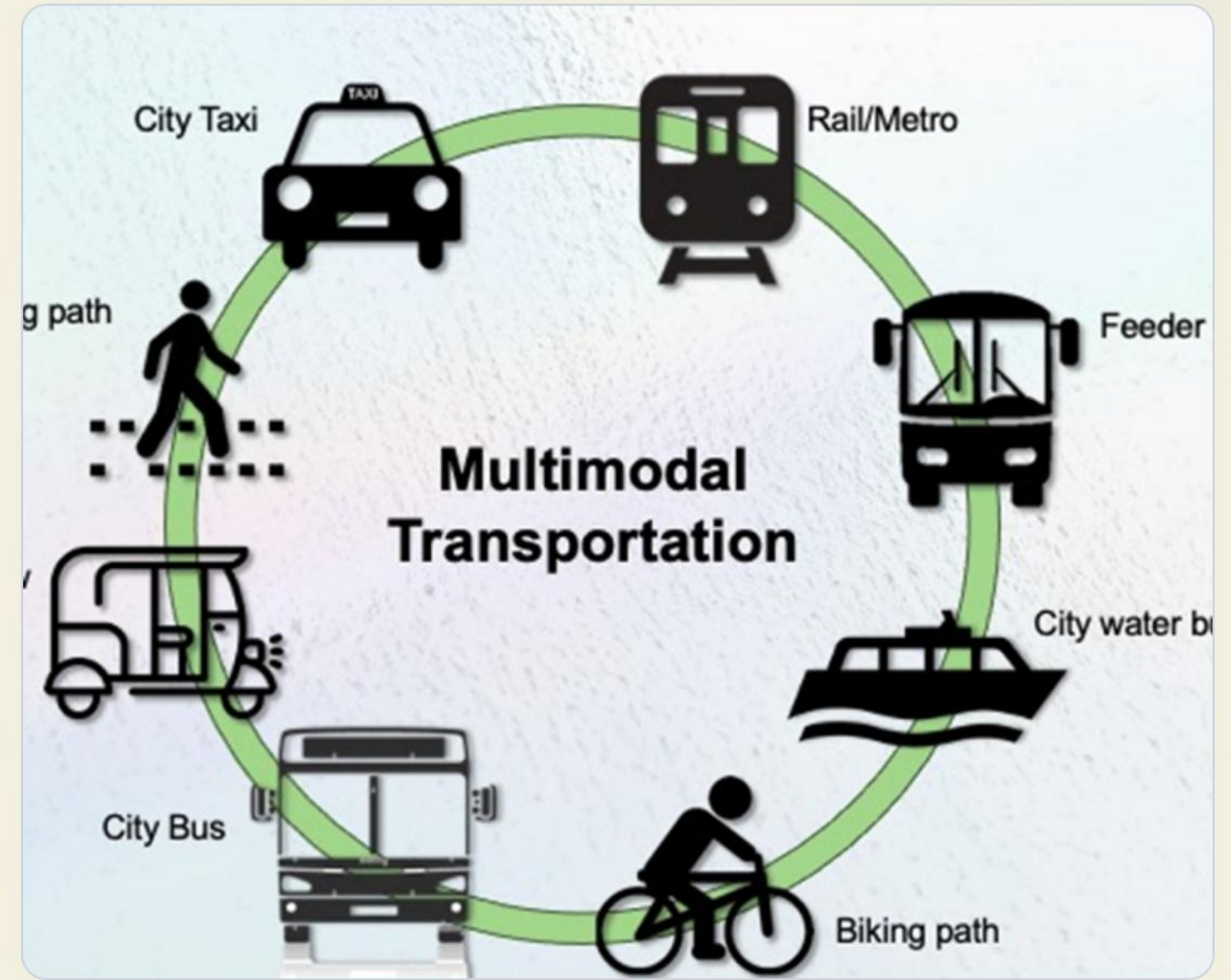
Supervisor: Mr. Yogiraj Bhale | Chandigarh University

EXECUTIVE SUMMARY

The Mobility Challenge

Indian metropolitan regions are inherently multi-modal. A single trip often combines walking, metro, bus, and last-mile rickshaws. However, existing navigation engines optimize only for time or distance.

P-E2R Framework: A time-dependent super-network approach using open data (OSM, GTFS, ERA5) to provide sustainable and personalized routing.



TARGET CLIENTS & STAKEHOLDERS



Commuters

Seeking reliable, low-delay travel and predictable transfers during peak city hours.



Students

Prioritizing low-fare options and affordability across varied transport modes.

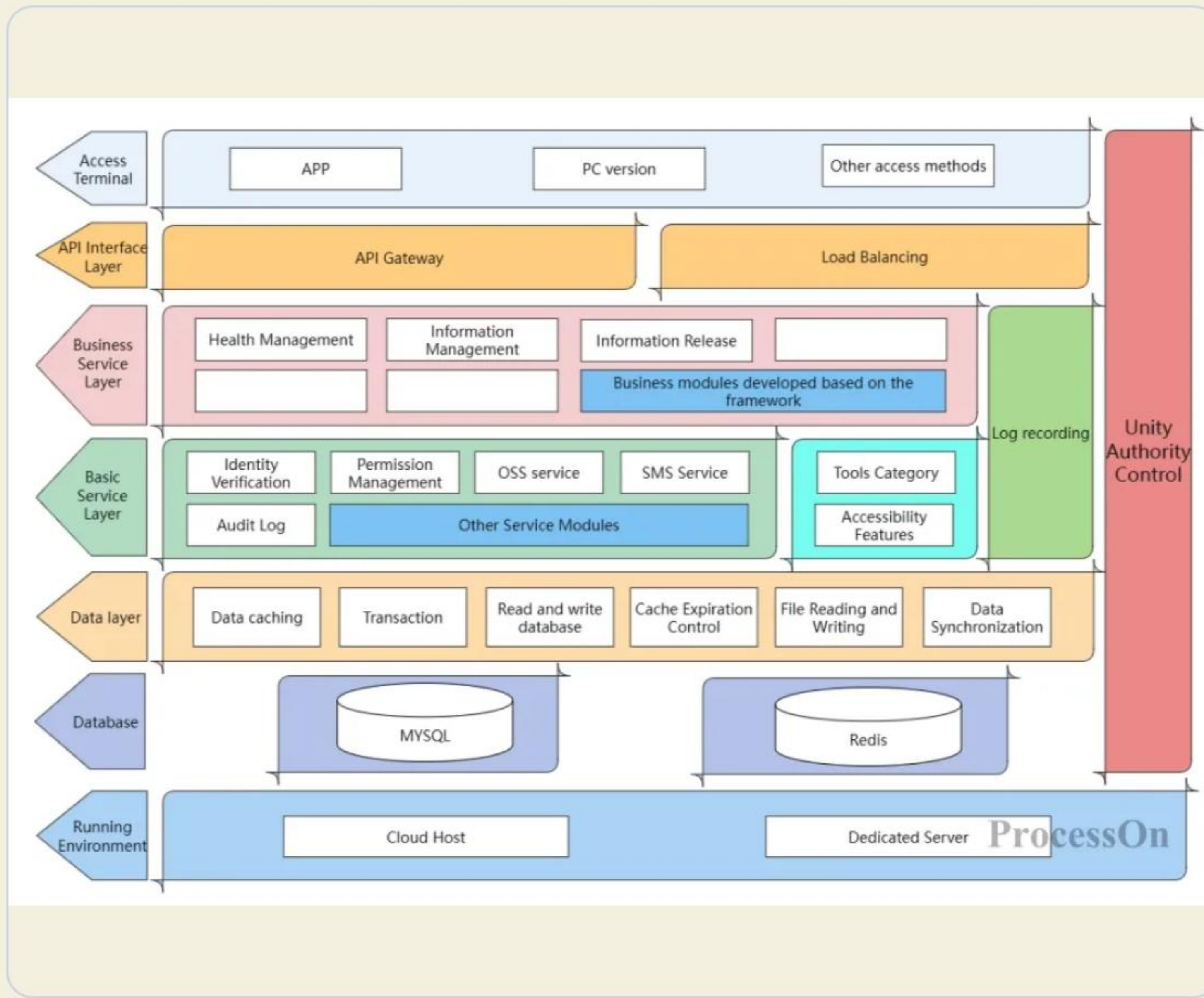


Health-Sensitive

Users aiming to minimize exposure to pollution hotspots and extreme urban heat.

"The framework balances operational efficiency with individual wellness and environmental sustainability."

P-E2R CORE ARCHITECTURE



 **Data Ingestion:** Normalizing OSM, GTFS, and environmental grids.

 **Feature Engineering:** Computing time, energy, and exposure proxies.

 **Optimization:** Pareto-efficient multi-criteria label-setting.

 **Personalization:** Ranking itineraries based on learned user weights.

Network Modeling & Optimization

Transforming fragmented transit data into a unified, time-dependent super-network
for seamless route planning.

MULTI-MODAL SUPER-NETWORK

Layered Representation

The city is modeled as a directed graph where nodes correspond to location-mode states. Walking, road-based (Rickshaws/Taxis), Bus, and Metro occupy distinct vertical layers connected by transfer edges.

Transfer Logic

Transfer edges explicitly model the friction of switching modes (waiting time, walking distance, fare barriers). This ensures that itineraries are not just "fast" on paper but practical in reality.

MATHEMATICAL FORMULATION

Each path $\mathbf{p}$ is evaluated through a multi-objective cost vector:

$$\mathbf{F}(\mathbf{p}) = [T(\mathbf{p}), C(\mathbf{p}), E(\mathbf{p}), G(\mathbf{p}), X(\mathbf{p}), D(\mathbf{p})]^T$$

Energy Model (E)

Proxies based on length, speed, road grade, and stop density.

Exposure Model (X)

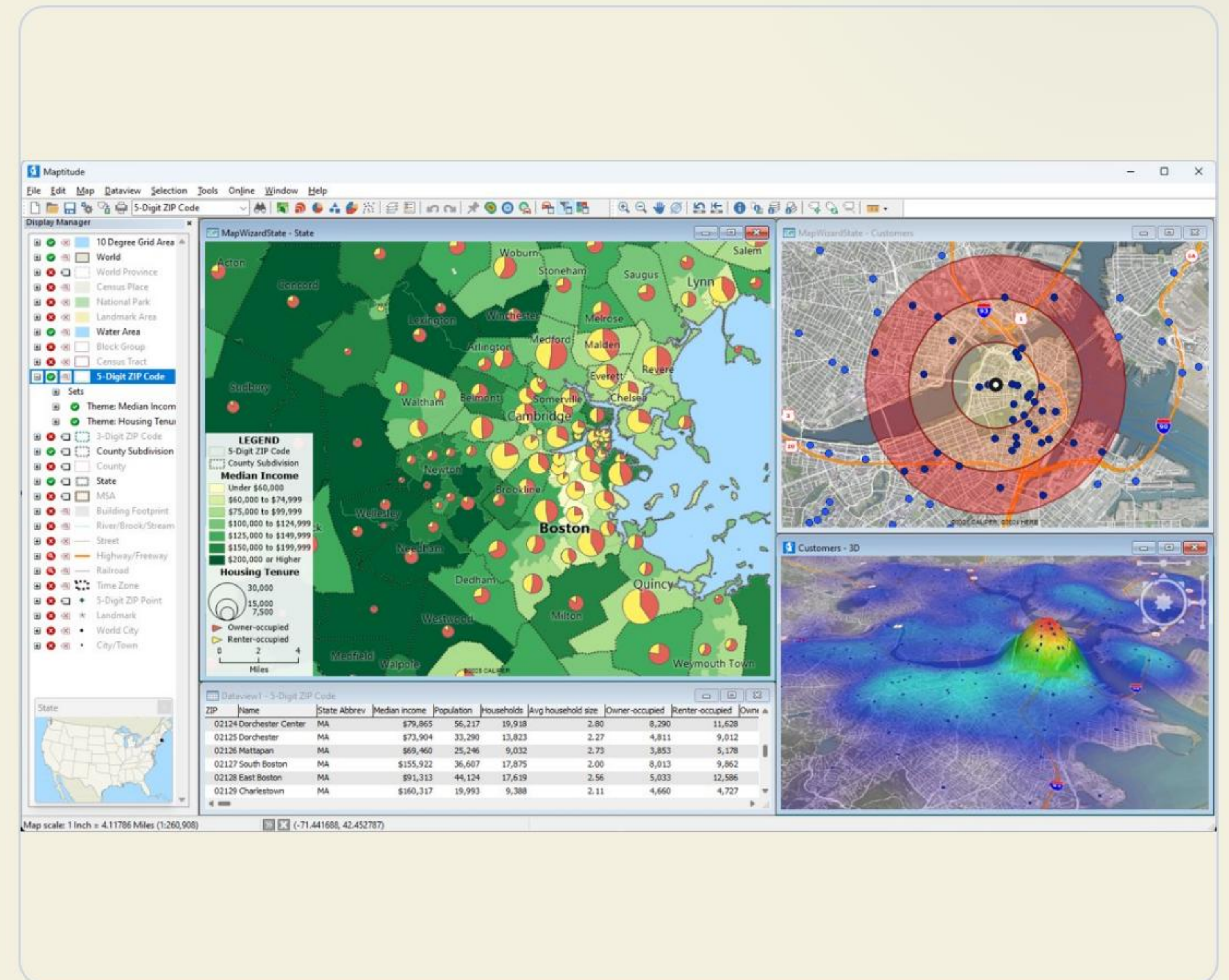
Aggregates PM2.5 concentration with mode-dependent multipliers.

DATA FUSION PIPELINE

Integrated Open Data

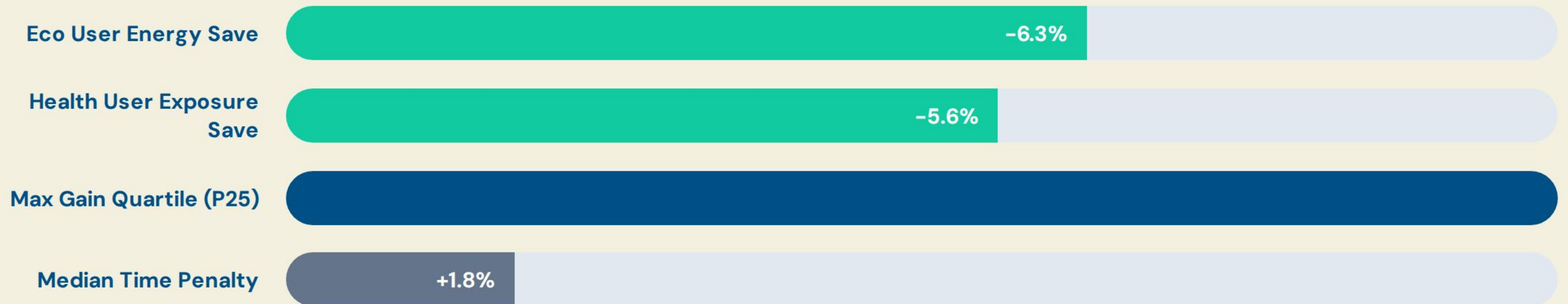
P-E2R is built on reproducible data streams to ensure adaptability across Indian Smart Cities:

- **OSM:** Base road and pedestrian graph.
- **GTFS/Flex:** Transit feeds & flexible IPT services.
- **CAMS/ERA5:** Global air quality and weather reanalysis.
- **DEM:** Digital elevation for energy/grade estimation.



QUANTITATIVE RESULTS SUMMARY

Simulation results across 250 OD pairs in Delhi, Bengaluru, and Mumbai show significant gains in secondary objectives with minimal time penalties.



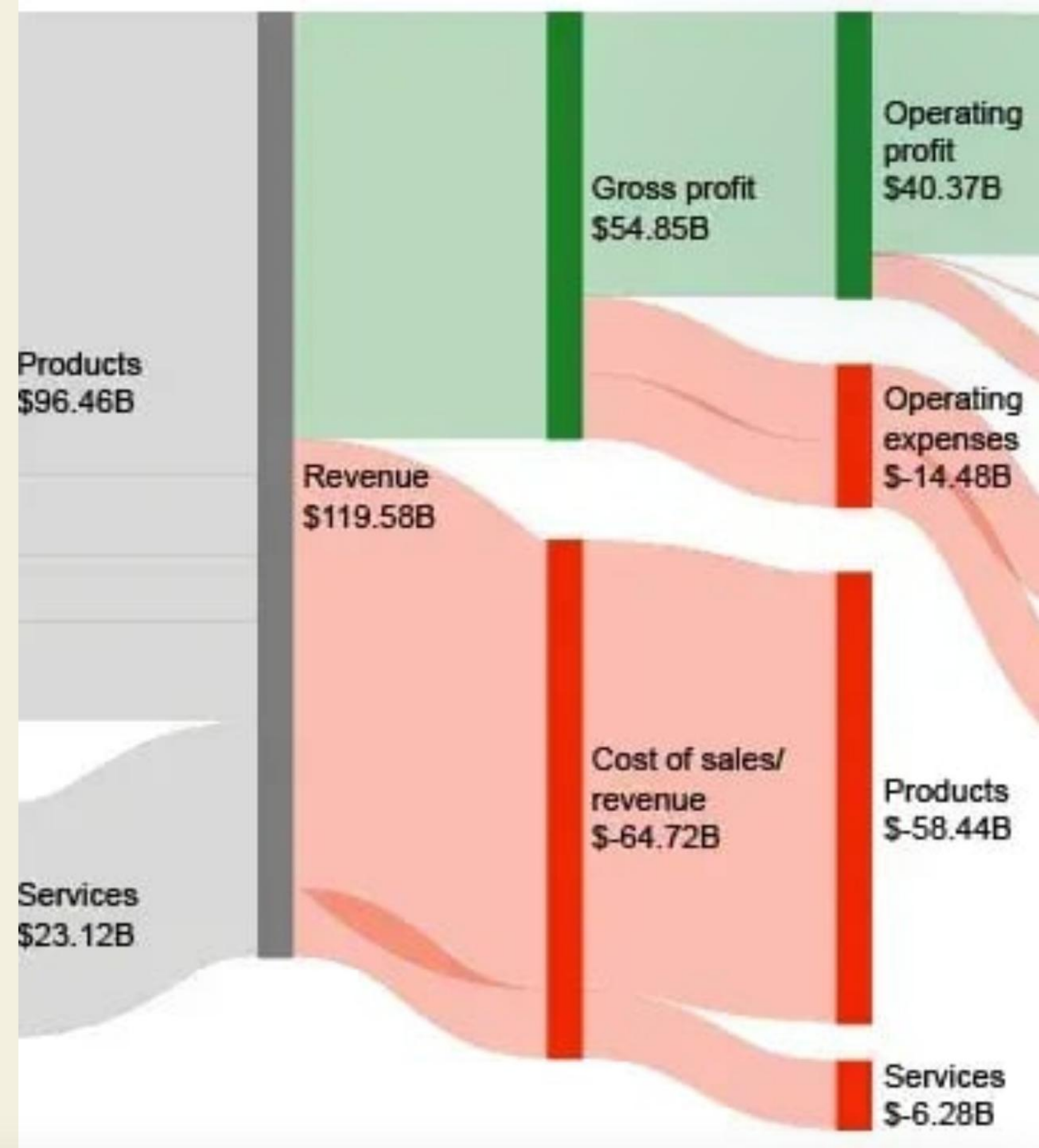
VALIDATION & TRUST

Uncertainty Reporting

Environmental sensors are sparse; P-E2R uses uncertainty bands and confidence messages to ensure honest route explanations.

User-Facing Logic

Explanation cards translate complex Pareto trade-offs into plain language: "+6 mins, -11% exposure".



CONCLUSION & ROADMAP

Prototype Ready

Modular architecture for personalized energy-aware routing using open data.

Pilot Scaling

Expansion to real-time GTFS integration and hyperlocal pollution sensors.

Operational MaaS

Enabling city agencies to meet sustainability goals through smarter routing.

P-E2R establishes that multi-modal, personalized routing is technically feasible and socially valuable for Indian smart cities.

Thank You

Questions?

Contact: soham.srivastava@university.in

Research Paper: P-E2R Framework for Indian Cities

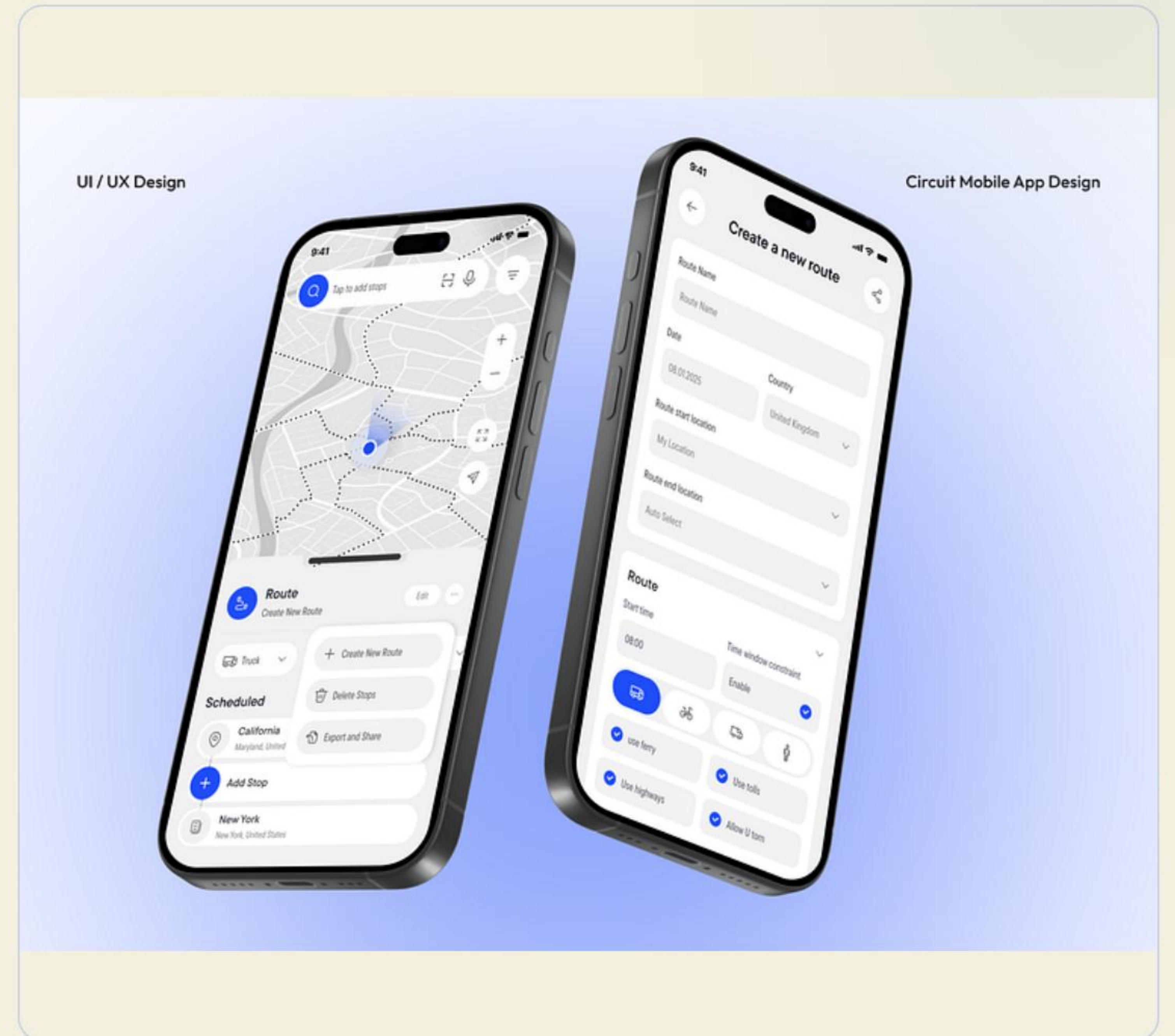
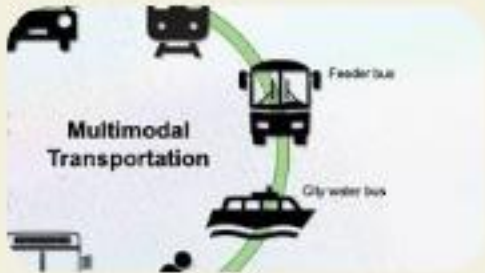


IMAGE SOURCES



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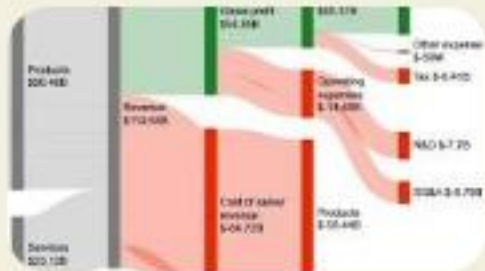
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